Course Code: 104

Course Title: Programming concepts and Programming methodology (PCPM)

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<tr><td>Course Code</td><td>104</td></tr>
<tr><td>Course Title</td><td>Programming Concepts and Programming Methodology (PCPM)</td></tr>
<tr><td>Credits</td><td>4</td></tr>
<tr><td>Course Category</td><td>Major Course</td></tr>
<tr><td>Level of Course</td><td>200-299 (Intermediate Level)</td></tr>
<tr><td>Teaching per Week</td><td>Theory/Lectures: 2 Hours/Week and Lab./Practical Hours: 4 Hours/Week</td></tr>
<tr><td>Minimum weeks per Semester</td><td>15 (Including class work, examination, preparation etc.)</td></tr>
<tr><td>Review / Revision</td><td>-</td></tr>
<tr><td>Implementation Year:</td><td>A.Y. 2023-2024</td></tr>
<tr><td>Purpose of Course</td><td>- Computer programming is a process that leads from an original formulation of a computing problem to executable computer programs.
- Programming involves activities such as analysis, developing, understanding, generating algorithms, verification of requirements of algorithms including their correctness, and implementation (commonly referred to as coding) of algorithms in a target programming language.
- To emphasis on concepts of Compiler based programming language, structure of code, algorithms, flow-charts, problem solving attitude, concepts of variables and declaration mechanism of different datatypes, simple I/O statements, conditional statements, loops, compound iterations, strings and certain inbuilt functions, header files, concepts of arrays and one dimensional numeric array operations, numeric inbuilt functions and concepts of pointers</td></tr>
<tr><td>Course Objective</td><td>To introduce students the essentials of computer Programming concepts and programming methodology using compiler based and interpreter based Programming languages.</td></tr>
<tr><td>Pre-requisite</td><td>-</td></tr>
<tr><td>Course Outcomes</td><td>CO1: Students will be able to learn fundamental programming concept of compiler based and interpreter based programming language.
CO2: Students will be proficient working on conditional statements, iterative Statements and fundamentals of programming concepts using C and Python.
CO3: Students will be able to represent compound data using lists, tuples and dictionaries in Python programs.
CO4: Students will be able to learn programming methodology.
CO5: Students will learn the concepts of interpreter based programming language using Python and its comparison with C programming language.</td></tr>
<tr><td>Mapping between Course Outcomes(CO) with Program Specific Outcomes(PSO)</td><td>
<table>
<tr><td></td><td>PSO1</td><td>PSO2</td><td>PSO3</td><td>PSO4</td><td>PSO5</td><td>PSO6</td><td>PSO7</td><td>PSO8</td></tr>
<tr><td>CO1</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>CO2</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>CO3</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>CO4</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>CO5</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
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<tr><td>Course Content</td><td>UNIT-1: Introduction
1.1 Concepts of Programming Languages
1.1.1 Introduction of Source Code, Object Code and executable code
1.1.2 Algorithm and Flowchart
1.1.3 Concepts of Structured Programming Language
1.2 Concepts and comparison of Interpreter and Compiler based languages:
1.2.1 Introduction of Interpreter and Compiler
1.2.2 Comparison of Interpreter and compiler
1.2.3 Introduction of compiler based(C language)program body structure
1.2.4 Introduction of interpreter based (Python) program body structure</td></tr>
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1.2.5 Character Set, concepts of variables and constants in C and Python
1.2.6 Identifiers, literals, Key words in C and Python
1.2.7 Data types in C (signed and unsigned) (Numeric : int, short int, long, float, double) , (Character type: char, string) and void.
1.2.8 Primitive Data types in Python
1.2.8 Concepts of source code, object code and executable code.

UNIT-2: Input/Output Statements and Operators (C Programming Language):

2.1 Input/Output statements:
2.1.1 Concepts of Header files (STDIO,CONIO)
2.1.1.1 Concepts of pre-compiler directives.
2.1.1.2 Use of #inlcude and #define
2.2 Input/Output Statements:
2.2.1 Input statements : scanf(), getc(), getch(), gets(), getchar()
2.2.2 Output Statements: printf(), putc(),puts(), putchar()
2.2.3 Type specifiers (formatting strings) : %d, %ld, %f, %c, %s, %lf
2.3 Operators :
2.3.1 Arithmetic operators (+, -, *, /, %, ++, --,)
2.3.2 Logical Operators (&&, ||, !)
2.3.3 Relational Operators (>, =, <=, !=)
2.3.4 Bit-wise operators (&, |, ^ , <>)
2.3.5 Assignment operators (=, +=, -=, *=, /=, %=)
2.3.6 Ternary Operator and use of sizeof() function.
2.4 Important Built-in functions:
2.4.1 Use of : (strlen, strcmp, strcpy, strcat,strrev)
2.4.2 Use of : (abs(), floor(), round(), ceil(), sqrt(), exp(), log(), sin(), cos(), tan(), pow() and trunc())

UNIT-3: Decision Making and Iterative statements in C programming Language:

3.1 if statements :
3.1.1 simple if statements
3.1.2 if…else statements
3.1.3 if…else if….else statements
3.1.4 Nested if statements.
3.2 Switch..case statements
3.2.1 Use of break and default
3.2.2 Difference between switch and if statements.
3.3 Use of goto statement for iteration
3.4 while loop
3.5 do..while loop
3.6 for loop
3.7 Nested while, do..while and for loops
3.8 Jumping statement: (break and continue)

UNIT-4: Concepts of Arrays and pointer in C Programming Language

4.1 Concepts of Single-dimensional Array
4.1.1 Numeric single dimensional Array
4.1.2 Numeric single dimensional array operations:
4.1.2.1 Sorting array in ascending or descending. (Bubble and selection)
4.1.2.2 Searching element from array (Linear Search)
4.1.3 Character Single dimensional Array
4.1.3.1 Character Single dimensional array operations:
4.1.3.2 Use of \0, \n and \t
4.2 Pointers:
4.2.1 Concepts of Pointers
4.2.2 Declaring and initializing int, float, char and void pointers
4.2.3 Pointer to single dimensional numeric array.

	UNIT-5: Concepts of Interpreter based programming language using Python 5.1 Concepts of Interpreter programming language 5.1.1 Start with Python code 5.1.2 Python Variables 5.1.3 Python Data structures 5.1.4 Data types and Data conversion in Python 5.2 Conditional and Iterative Statements 5.2.1 Conditional Statements 5.2.2 Iterative Statements 5.2.3 Python Operators 5.3 Python In-built collections 5.3.1 List 5.3.2 Tuples 5.3.3 Sets 5.3.4 Dictionary [Practical work for Unit-1 to 4 will be carried out for 'C' Programming Language and Unit-5 for 'Python Programming language'.]
Reference Books	1. Programming in C, Balaguruswami – TMH 2. C: How to Program, Deitel & Deitel - PHI 3. C Programming Language, Kernigham & Ritchie - TMH 4. Programming in C, Stephan Kochan - CBS 5. Mastering Turbo C, Kelly & Bootle - BPB 6. C Language Programming – Byron Gottfried - TMH 7. Let us C, Yashwant Kanetkar - BPB Publication 8. Magnifying C, Arpita Gopal - PHI 9. Problem Solving with C, Somashekara - PHI 10.Programming in C, Pradip Dey & Manas Ghosh – Oxford 11. "Learning Python" by Mark Lutz, ISBN: 978-1449355739, Publisher: O'Reilly Media 12. "Python for Data Analysis" by Wes McKinney, ISBN: 978-1491957660, Publisher: O'Reilly Media
Teaching Methodology	Class Work, Discussion, Lab work, Self-Study, Seminars and/or Assignments
Evaluation Method	50% Internal assessment. 50% External assessment.